Ashutosh Sharma

Cloud Engineering Intern Candidate · AWS · Terraform · Python · Shell Scripting

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SUMMARY

Aspiring cloud engineer seeking a cloud engineering internship, with hands-on experience building serverless systems on AWS using infrastructure as code. Built and deployed LiftLens on Lambda, DynamoDB and Cognito, with every resource in Terraform and cost guardrails that hold it at $0/month. Independent research on cloud workload forecasting: replicated a 2025 FGCS method on the Alibaba 2018 cluster trace and found flaws in its evaluation metrics.

EDUCATION

B.Tech, Information Technology — Netaji Subhas University of Technology (NSUT), New Delhi · Expected 2028
Class XII, High School Diploma — American International School of Mozambique (AISM), Maputo · 2023 · 89%
Class X, IB MYP — American International School of Mozambique (AISM), Maputo · 2021 · 82%

EXPERIENCE

Digital Marketing Intern — Socialbent, Gurgaon · May 2025 – Jul 2025

- Supported SEO research and content optimization for client campaigns.
- Contributed to social media content planning and campaign execution.
- Tracked digital campaign performance with the marketing team as part of campaign management.
- Collaborated in a professional team environment while meeting project deadlines.

RESEARCH

Re-evaluation of spike-aware cloud CPU forecasting — Independent research · Sep 2026 – Present

- Critically re-evaluated a 2025 *Future Generation Computer Systems* (FGCS) paper on spike-aware cloud CPU forecasting and identified two flaws in its published evaluation, both confirmed in the authors' code: a numerically degenerate over-provisioning metric and a precision metric that rewards lag.
- Replicated the method on the Alibaba 2018 cluster trace (29 GB, ~6k containers), processing it with Python (pandas) and analysing the results in Excel with formulas and charts.
- Showed that a retraining-free quantile readout of a standard classifier matches the method's spike recall with ~30% less over-provisioning, and that the method's advantage appears only under actuation delay.

PROJECTS

LiftLens — privacy-first physique tracker on AWS · React, TypeScript, Python, Lambda, DynamoDB, Cognito, Terraform · GitHub

- Built a 3-tier serverless web app (React/TypeScript, Python on AWS Lambda, DynamoDB, Cognito) managed entirely with Terraform (25 resources across three stacks), running at $0 a month.
- Implemented on-device pose estimation with MediaPipe, so body photos are never uploaded or stored.
- Designed an explainable score: percent change against the user's own baseline, with a noise band calibrated from repeat photos (±2.4%).
- Engineered photo-quality checks with thresholds tested on real photos, so bad photos get a retake message instead of a misleading number.
- Built cost guardrails, tested end to end: a $1 budget alert, a kill-switch Lambda and a budget action that locks deploys.
- Secured the API with JWT verification against Cognito, and covered it with 89 TypeScript and 112 Python tests, including forged-signature and expired-token cases.
- Automated builds and configuration with Bash scripts that cross-build the Lambda package for Linux arm64 and generate frontend config from Terraform outputs.
- Documented every major decision as an ADR (8 so far), with the alternatives considered and the trade-offs.

LEADERSHIP

Student Lead & Organizer — School Clothing Drive (IB CAS) · Nov 2022 – Dec 2022

- Led a team of peer volunteers to organize and run a school-wide clothing drive benefiting local youth communities.
- Coordinated logistics and distribution with the school administration so every donation was delivered safely to the target communities.

SKILLS

Cloud (AWS): Lambda, DynamoDB, Cognito, CloudWatch, SNS, Budgets, IAM and IAM Identity Center (SSO)
Infrastructure as code: Terraform (multi-stack layout, plan/apply review, deletion protection, tag-based resource targeting)
Security and cost: least-privilege deploy role, JWT verification, budget alerts and automated kill switch
Languages and data: Python (pandas), TypeScript, JavaScript, Java, SQL, HTML, CSS, Microsoft Excel (formulas, charts)
Frontend: React, Vite, MediaPipe (in-browser pose detection)
Practices: automated testing (Vitest, pytest), architecture decision records, Git and pull-request workflow

CERTIFICATIONS

AWS Certified Cloud Practitioner — in preparation · AWS Certified Solutions Architect – Associate — planned